Supplementary Appendix

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This appendix has been provided by the authors to give readers additional information about the work.

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Inclusion and Exclusion Criteria

Healthy participants with pre-existing stable disease could be included, including those with chronic stable HIV, hepatitis C virus, or hepatitis B virus infection. Exclusion criteria included previous clinical or virologic COVID-19 diagnosis, receipt of COVID-19 preventative treatments, history of severe adverse reaction with a vaccine or any component of the study intervention, and immunodeficiency, autoimmune disease, or conditions associated with prolonged bleeding.

Ethical Conduct of the Studies

Conduct of the study was in accordance with the study protocol and with consensus ethical principles derived from international guidelines including the Declaration of Helsinki, Council for International Organizations of Medical Sciences Ethical Guidelines, International Conference on Harmonisation Guidelines for Good Clinical Practice, and applicable laws and regulations, including applicable privacy laws. The study protocols and any amendments, informed consent documents, and other relevant documents were approved by institutional review board/ethics committees before study initiation. Before any study activity, written informed consent was obtained.

Determination of SARS-CoV-2 Infection and COVID-19 Illness

Surveillance for potential cases of COVID-19 occurred throughout a participant's involvement in the study. If a participant developed an acute respiratory illness, the participant was considered to potentially have COVID-19 illness. In this circumstance, assessments were to include nasal (mid-turbinate) swab sample collection, which was tested at a central laboratory using the reverse transcription–polymerase chain reaction (RT-PCR) test (Cepheid) or other equivalent nucleic acid amplification–based test (NAAT), to detect SARS-CoV-2. Clinical information and results from local standard-of-care tests were also assessed. The central laboratory NAAT result was used for the case definition. If no result was available from the central laboratory, a local NAAT result could be used if it was obtained using either the Cepheid Xpert Xpress SARS-CoV-2, Roche Cobas SARS-CoV-2 Real-Time RT-PCR test, or the Abbott Molecular/RealTime SARS-CoV-2 assay.

SARS-CoV-2–related cases and SARS-CoV-2–related severe cases were documented. For both, the onset date of the case was the date that symptoms were first experienced by the participant; if new symptoms were reported within 4 days after resolution of all previous symptoms, they were considered as part of a single illness. Participants used an electronic diary to record symptoms. The definition of confirmed COVID-19 included the presence of at least one symptom (ie, fever, new or increased cough, new or increased shortness of breath, chills, new or increased muscle pain, new loss

of taste or smell, sore throat, diarrhea, vomiting) and being SARS-CoV-2 NAAT-positive during, or within 4 days before or after, the symptomatic period (either at the central laboratory or at a local testing facility using an acceptable test).

Diagnosis of severe COVID-19 according to the US Food and Drug Administration definition (www.fda.gov/media/139638/download) included confirmed COVID-19 and the presence of at least one of the following: clinical signs at rest indicative of severe systemic illness (eg, respiratory rate ≥30 breaths per minute, heart rate ≥125 beats per minute, SpO_2 ≤93% on room air at sea level, or PaO_2/FiO_2 <300 mmHg); respiratory failure (defined as needing high-flow oxygen, non-invasive ventilation, mechanical ventilation, or extracorporeal membrane oxygenation); evidence of shock (ie, systolic blood pressure <90 mm Hg, diastolic blood pressure <60 mm Hg, or requiring vasopressors); significant acute renal, hepatic, or neurologic dysfunction; intensive care unit admission; or death.

Diagnosis of severe COVID-19 according to the US Centers for Disease Control and Prevention definition (https://www.cdc.gov/coronavirus/2019-ncov/need-extra-precautions/people-with-medical-conditions.htm) included confirmed COVID-19 and the presence of at least one of the following: hospitalization, ICU admission, intubation or mechanical ventilation, or death.

Determination of Relative Vaccine Efficacy

At the planned interim analysis, relative vaccine efficacy of the BNT162b2 dose 3 group compared to the placebo group (participants who received the primary series of 2 BNT162b2 doses ≥6 months before enrollment in the study and who did not receive BNT162b2 dose 3) was determined.

Relative vaccine efficacy was estimated as 100 × (1–IRR), where IRR is the calculated ratio of confirmed COVID-19 illness per 1000 person-years of follow-up in the BNT162b2 dose 3 group to the corresponding illness rate in the placebo dose 3 group.

Table S1. Study populations

Population	Description
Enrolled	All participants who provided informed consent
Randomized	All participants who were assigned a randomization number
Evaluable efficacy*	All eligible randomized participants who received dose 3 vaccination as randomized and had no other important protocol deviations
All-available efficacy†	All randomized participants who receive at least one dose of the study intervention
Safety	All randomized participants who receive at least one dose of the study intervention

* The efficacy population with complete data, which is the primary efficacy analysis population.
† The total efficacy population.

Table S2. Representativeness of study participants*

Category	Details
Disease, problem, or condition under investigation	COVID-19
Special considerations related to	
Sex and gender	COVID-19 rates are similar between men and women, although men appear at increased risk for severe disease including hospitalization, intensive care admission, and death.[1] Limited data are available on the relationship between gender identity and COVID-19.[2]
Age	COVID-19 cases are generally of similar prevalence across adult age groups.[3] However, rates of hospitalization and death increase with older age.
Race or ethnic group	All races and ethnicities are affected by COVID-19.[4] Racial and ethnic minority populations are at increased risk for COVID-19 hospitalization and mortality.[5] For instance, in the United States, Black/African American, Hispanic/Latinx, and American Indian/Alaska Native individuals have higher rates of hospitalization and death compared with White individuals.[4]
Geography	COVID-19 cases have been reported worldwide, although the number of cases and outcomes have varied by country/region.[6] As of January 26, 2022, the number of COVID-19 cases and attributed deaths in the United States were >71 million and >862,000, respectively. Corresponding values are >3.5 million and >94,000 for South Africa and >24 million and >623,000 for Brazil.
Other considerations	Several medical conditions and comorbidities increase the risk of developing severe COVID-19 (eg, including obesity, diabetes, and chronic pulmonary disease), and the risk increases as the number of underlying conditions increases.[7]
Overall representativeness of this trial	The participants included in this study were 16 years and older. The median age of participants was 53 years and participants ≥65 years old were somewhat underrepresented in the trial (23% of the study population). The proportion of men and women in the trial was similar. The trial disproportionately included White participants (79%), with 9%, 6%, and 15% of participants of Black/African American, Asian, and Hispanic/Latinx race/ethnicity. Overall, 48% of participants had comorbidities. The study was conducted in the United States, South Africa, and Brazil, which have reported high numbers of COVID-19 cases and deaths.

1. Tharakan et al. Nat Rev Urol. 2021;19:47-63.
2. Morgan et al. Front Sociol. 2021;6:650729.
3. US Centers for Disease Control and Prevention. Risk for COVID-19 infection, hospitalization, and death by age group. Updated November 22, 2021. https://www.cdc.gov/coronavirus/2019-ncov/covid-data/investigations-discovery/hospitalization-death-by-age.html. Accessed January 27, 2022.
4. US Centers for Disease Control and Prevention. Risk for COVID-19 infection, hospitalization, and death by race/ethnicity. Updated November 22, 2021. https://www.cdc.gov/coronavirus/2019-ncov/covid-data/investigations-discovery/hospitalization-death-by-race-ethnicity.html. Accessed January 27, 2022.
5. Khanijahani et al. Int J Equity Health. 2021;20:248.
6. World Health Organization. WHO Coronavirus (COVID-19) Dashboard. https://covid19.who.int/. Accessed January 27, 2022.
7. US Centers for Disease Control and Prevention. Underlying medical conditions associated with higher risk for severe COVID-19: information for healthcare providers. https://www.cdc.gov/coronavirus/2019-ncov/hcp/clinical-care/underlyingconditions.html#:~:text=Main%20Findings,the%20strongest%20association%20with%20death. Accessed January 27, 2022.

* A literature search of PubMed and government sources was conducted to determine the representativeness of the participant population to the population of individuals with COVID-19. Where applicable, the focus of the search was adult populations in the United States, South Africa, and Brazil, which are the populations included in this clinical study.

Table S3. Serious adverse events by system organ class and preferred term

	BNT162b2 (N*=5055)	Placebo (N*=5020)
Serious adverse events from third dose to cutoff date, n† (%)	16 (0.3)	24 (0.5)
Cardiac disorders	2 (0.0)	3 (0.1)
Acute myocardial infarction	1 (0.0)	1 (0.0)
Myocardial infarction	0	1 (0.0)
Tachycardia	1 (0.0)	0
Ventricular extrasystoles	0	1 (0.0)
Gastrointestinal disorders	1 (0.0)	0
Intestinal obstruction	1 (0.0)	0
General disorders and administration site conditions	0	2 (0.0)
Chest pain	0	2 (0.0)
Infections and infestations	4 (0.1)	4 (0.1)
Appendicitis	1 (0.0)	0
Appendicitis perforated	1 (0.0)	0
COVID-19 pneumonia	0	1 (0.0)
Diverticulitis	1 (0.0)	0
Pneumonia	0	1 (0.0)
Salmonellosis	1 (0.0)	0
Sepsis	0	1 (0.0)
Urinary tract infection	0	1 (0.0)
Injury, poisoning and procedural complications	1 (0.0)	2 (0.0)
Acetabulum fracture	1 (0.0)	0
Hip fracture	0	1 (0.0)
Pelvic fracture	1 (0.0)	0
Thoracic vertebral fracture	0	1 (0.0)
Investigations	2 (0.0)	0
Hepatic enzyme increased	2 (0.0)	0
Musculoskeletal and connective tissue disorders	1 (0.0)	1 (0.0)
Intervertebral disc protrusion	1 (0.0)	0
Osteoarthritis	0	1 (0.0)
Neoplasms benign, malignant and unspecified (including cysts and polyps)	2 (0.0)	5 (0.1)
Follicular lymphoma	1 (0.0)	0
Lung carcinoma cell type unspecified stage II	0	1 (0.0)
Metastases to diaphragm	0	1 (0.0)
Metastases to liver	0	1 (0.0)
Pancreatic carcinoma	0	1 (0.0)
Pancreatic carcinoma metastatic	0	1 (0.0)
Prostate cancer	0	1 (0.0)
Renal cancer metastatic	0	1 (0.0)
Renal cell carcinoma	1 (0.0)	0
Nervous system disorders	3 (0.1)	2 (0.0)
Cerebrovascular accident	1 (0.0)	1 (0.0)
Cerebral venous thrombosis	0	1 (0.0)
Syncope	1 (0.0)	0
Toxic encephalopathy	1 (0.0)	0
Pregnancy, puerperium and perinatal conditions	0	1 (0.0)
Abortion spontaneous	0	1 (0.0)
Psychiatric disorders	1 (0.0)	0
Suicidal ideation	1 (0.0)	0

Renal and urinary disorders	1 (0.0)	0
Nephrolithiasis	1 (0.0)	0
Reproductive system and breast disorders	0	1 (0.0)
Adenomyosis	0	1 (0.0)
Respiratory, thoracic and mediastinal disorders	0	6 (0.1)
Pulmonary embolism	0	3 (0.1)
Chronic obstructive pulmonary disease	0	1 (0.0)
Dyspnea	0	1 (0.0)
Respiratory failure	0	1 (0.0)
Vascular disorders	0	1 (0.0)
Hypertension	0	1 (0.0)

Results are for the safety population (**Table S1**); participants with stable HIV were analyzed separately. The cutoff date was October 5, 2021.

* Number of participants in the specified group. This value is the denominator for the percentage calculations.

† Number of participants reporting ≥1 occurrence of the specified event.

Table S4. Relative vaccine efficacy against first COVID-19 occurrence from 7 days after dose 3 by subgroup in participants without evidence of prior SARS-CoV-2 infection

	BNT162b2 (N=4695)		Placebo (N=4671)		RVE (95% CI§)
	n1*	Surveillance time† (n2‡)	n1*	Surveillance time† (n2‡)	
Overall	6	0.823 (4659)	123	0.792 (4614)	95.3% (89.5, 98.3)
Age group					
16–55 years	3	0.446 (2548)	81	0.427 (2523)	96.5% (89.3, 99.3)
>55 years	3	0.378 (2111)	42	0.365 (2091)	93.1% (78.4, 98.6)
16–30 years	0	0.080 (457)	11	0.079 (466)	100.0% (60.5, 100.0)
31–40 years	0	0.124 (711)	31	0.115 (682)	100.0% (88.3, 100.0)
41–50 years	2	0.152 (876)	21	0.153 (901)	90.5% (60.9, 98.9)
51–60 years	2	0.185 (1034)	36	0.170 (990)	94.9% (80.1, 99.4)
>60 years	2	0.282 (1581)	24	0.275 (1575)	91.9% (67.3, 99.1)
≥75 years	0	0.045 (249)	1	0.042 (241)	100.0% (–3607.9, 100.0)
Sex					
Male	4	0.398 (2258)	70	0.396 (2310)	94.3% (84.8, 98.5)
Female	2	0.425 (2401)	53	0.396 (2304)	96.5% (86.7, 99.6)
Race					
White	5	0.661 (3727)	100	0.638 (3709)	95.2% (88.4, 98.5)
Black/African American	0	0.064 (363)	13	0.063 (364)	100.0% (68.0, 100.0)
American Indian or Alaska Native	0	0.013 (80)	4	0.013 (86)	100.0% (–59.2, 100.0)
Asian	1	0.046 (272)	3	0.043 (255)	69.4% (–280.7, 99.4)
Multiracial	0	0.034 (188)	2	0.031 (175)	100.0% (–395.3, 100.0)
Not reported	0	0.004 (22)	1	0.002 (14)	100.0% (–2233.3, 100.0)
Ethnicity					
Hispanic/Latinx	1	0.118 (683)	19	0.116 (683)	94.8% (67.5, 99.9)
Non-Hispanic/non-Latinx	5	0.704 (3967)	104	0.675 (3923)	95.4% (88.9, 98.5)
Country					
Brazil	0	0.092 (518)	4	0.093 (526)	100.0% (–52.7, 100.0)
South Africa	0	0.012 (74)	2	0.015 (93)	100.0% (–554.9, 100.0)
United States	6	0.719 (4067)	117	0.684 (3995)	95.1% (89.1, 98.2)
Time between dose 2 and dose 3					
6–<8 months	1	0.115 (698)	10	0.109 (676)	90.5% (33.4, 99.8)
8–<10 months	1	0.132 (738)	26	0.128 (741)	96.3% (77.3, 99.9)
10–<12 months	4	0.550 (3058)	82	0.531 (3037)	95.3% (87.5, 98.7)
≥12 months	0	0.027 (165)	5	0.024 (160)	100.0% (–0.2, 100.0)

Results are for the evaluable efficacy population (**Table S1**). Participants without evidence of infection had no serological or virological evidence (before 7 days after receipt of dose 3) of past SARS-CoV-2 infection (ie, N-binding antibody [serum] negative at vaccination visit 1, SARS-CoV-2 not detected in nasal swabs by NAAT at vaccination visit 1, and negative NAAT at any unscheduled visit before 7 days after dose 3).

* Number of participants meeting the endpoint definition.

† Total surveillance time in 1000 person-years for the given endpoint across all participants within each group at risk for the endpoint. Time period for COVID-19 case accrual is from 7 days after dose 3 to the end of the surveillance period.

‡ Number of participants at risk for the endpoint.
§ 95% CI for RVE is derived based on the Clopper–Pearson method adjusted for surveillance time. The lower limit of the CI could be negative if the point estimate was low or if the CI was wide due to a small sample size.
NAAT=nucleic acid amplification test; RVE=relative vaccine efficacy of the BNT162b2 dose 3 group relative to the placebo group.

Table S5. Relative vaccine efficacy for first COVID-19 occurrence from 7 days after dose 3 by comorbidity status in participants without evidence of prior SARS-CoV-2 infection

	BNT162b2 (N=4695)		Placebo (N=4671)		RVE (95% CI§)
	n1*	Surveillance time† (n2‡)	n1*	Surveillance time† (n2‡)	
Overall	6	0.823 (4659)	123	0.792 (4614)	95.3% (89.5, 98.3)
Comorbidity					
No comorbidity	1	0.429 (2426)	59	0.406 (2383)	98.4% (90.7, 100.0)
Any comorbidity	5	0.394 (2233)	64	0.385 (2231)	92.4% (81.2, 97.6)
Any malignancy	1	0.044 (247)	4	0.037 (214)	78.6% (–116.4, 99.6)
Cardiovascular	2	0.030 (169)	1	0.023 (133)	–57.3% (–9181.8, 91.8)
Chronic pulmonary disease	1	0.071 (404)	16	0.074 (437)	93.4% (57.8, 99.8)
Diabetes	1	0.068 (386)	17	0.066 (384)	94.3% (63.3, 99.9)
Obese (BMI ≥30 kg/m^2)	2	0.289 (1634)	52	0.283 (1637)	96.2% (85.7, 99.6)
Hypertension	3	0.226 (1266)	33	0.221 (1265)	91.1% (71.7, 98.3)
Diabetes (including gestational diabetes)	1	0.068 (389)	17	0.067 (387)	94.3% (63.3, 99.9)

Results are for the evaluable efficacy population (**Table S1**). Participants without evidence of infection had no serological or virological evidence (before 7 days after dose 3) of past SARS-CoV-2 infection (ie, N-binding antibody [serum] negative at vaccination visit 1, SARS-CoV-2 not detected in nasal swabs by NAAT at vaccination visit 1, and negative NAAT at any unscheduled visit before 7 days after dose 3). Comorbid conditions are from the Charlson Comorbidity Index or BMI ≥30 kg/m^2.

* Number of participants meeting the endpoint definition.

† Total surveillance time in 1000 person-years for the given endpoint across all participants within each group at risk for the endpoint. Time period for COVID-19 case accrual is from 7 days after dose 3 to the end of the surveillance period.

‡ Number of participants at risk for the endpoint.

§ 95% CI for RVE is derived based on the Clopper–Pearson method adjusted for surveillance time. The lower limit of the CI could be negative if the point estimate was low or if the CI was wide due to a small sample size.

BMI=body mass index; NAAT=nucleic acid amplification test; RVE=relative vaccine efficacy of the BNT162b2 dose 3 group relative to the placebo group.